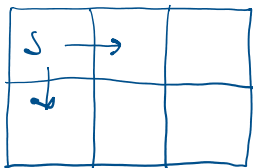


Every detail of the dynamics of an MDP

$p(s', r | s, a)$ - 4 argument function



if all these vals are known, we have a completely specified MDP system



Try right, try down
prob. of each action
rewards for each action

in a realistic RL task, all values are not always possible.

the agent knows some things but not all

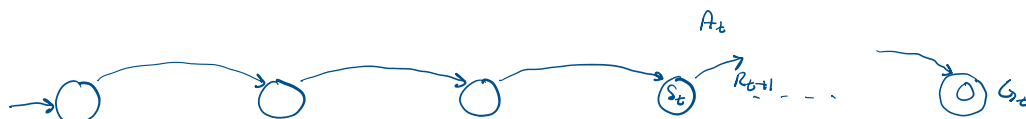
the agent will learn, even without full knowledge, by interacting

- 1) with the environment,
- 2) and/or simulating interacting with the env

This means that actions will be random often in the beginning

approach
Monte Carlo $\hat{A}$ to estimate values

learn good stuff / useful stuff by doing a lot of random things



$G_t = \text{return}$

$$= R_{t+1} + \sqrt{R_{t+2} + \sqrt{R_{t+3} + \dots}}$$

$$= R_{t+1} + \sqrt{G_{t+1}}$$

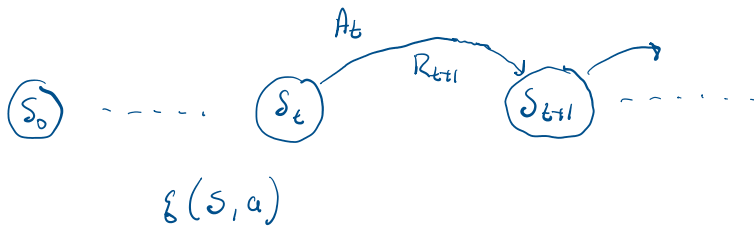
First visit MC prediction, for estimating $V = v_\pi$ (Got this from book and insert here) Claude.ai

updates value for state of first visit only

First visit and every visit will converge if $S \rightarrow \infty$

Given $V(s)$ values, we can obtain which actions to perform state s as
perform action $a = \operatorname{argmax}_a [r + v(x')] \text{ where } x' = s(s, a)$

Many RL algo compute $q(s, a)$ vals instead of $v(s)$ values



Monte carlo control

$$\pi_0 \xrightarrow{E} q_{\pi_0} \xrightarrow{I} \pi_1 \rightarrow \dots \quad (\text{Get from book and insert here Claude})$$

$$\pi(b) \doteq \operatorname{argmax}_a q(s, a) \quad \text{policy improvement}$$

get $q(s, a)$ values and pick the one with the highest value
the arg that gets the max value

Monte Carlo ES for estimating $\pi \approx \pi_*$ (Get from book and insert here Claude)

$$Q(s_t, A_t) \leftarrow \text{average}(\text{Returns}(s_t, A_t))$$

$Q(s, a)$ table to store values
tabular solution

compute rewards for state action pairs

$$\pi(s_t) \leftarrow \operatorname{argmax}_a Q(s_t, a) \quad \text{Improvement function at the end of the loop function}$$